Fiscal Year:	FY 2005	**Task Last Updated:**	FY 09/19/2005
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Project Title:	Radiation Shielding Properties of Multifunctional Spacecraft Materials		
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Program/Discipline:	RADIATION SHIELDING		
Program/Discipline--Element/Subdiscipline:	RADIATION SHIELDING--Radiation shielding		
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Space Biology Special Category:	None		
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Zip Code:	94915-0788	**Congressional District:**	3
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Task Description:	MMARSS: Multifunctional Materials Analysis of Radiation Shielding for Spacecraft is an extensive, systematic program to measure and characterize the space radiation shielding properties of a wide range of potential multifunctional spacecraft materials. The program contains both a materials fabrication component and a laboratory testing component wherein the shielding properties of candidate materials are measured at ground-based particle accelerators. Laboratory testing involves the measurement of the radiation field behind samples of the various shielding materials during heavy ion exposures using the NASA Space Radiation Laboratory (NSRL) at the Brookhaven National Laboratory (BNL) and the HIMAC heavy ion accelerator at the National Institute for Radiological Sciences (NIRS), Chiba, Japan. The radiation shielding properties of candidate multifunctional materails are measured as a function of shielding thickness using a suite of radiation detectors including CR-39 Plastic Nuclear Track Detectors (PNTDs), Tissue Equivalent Proportional Counters (TEPCs), Thermoluminescent Detectors (TLDs) and Si spectrometers. Measured quantities include LET spectra and charge spectra from CR-39 PNTDs and Si spectrometers, Lineal Energy Spectra from the TEPCs, and dose, dose equivalent and average quality factor as a function of shielding thickness from all instruments. Multifunctional spacecraft materials are selected on the basis of their potential use as space radiation shielding, i.e. materials with low mean atomic mass and high hydrogen content, and also on the basis of their suitability for use in aerospace vehicle structures. The concentration during the first year of this project has been on the testing of composite materials including carbon and polyethylene fiber composites. 1. Selection and fabrication of multifunctional spacecraft material sample targets: Candidate multifunctional materials are chosen on the basis of their potential use as shielding from space radiation including GCR and solar protons and use as structural, thermal protection, micrometeriod protection or other subsystems of the spacecraft. In terms of radiation shielding, materials of low mean atomic mass and high hydrogen content are favored. In terms of spacecraft structures, light weight materials with good mechanical and thermal properties are sought. Materials with a high NASA Technical Readiness Level (TRL) such as Carbon and Polymer Composites, Structural Polymers, and Metal Hydrides, are being given a high priority. One type of material currently with a low TRL, Carbon Nanotubes, is included in the program due to the tremendous benefits that could be derived from such materials both structurally and in terms of radiation shielding, should these materials be successfully developed. In addition to single materials, layers of different materials with differing radiation shielding properties are also being tested, as are materials based on the use of in situ resources, e.g. Lunar and Martian regolith. The selection of materials for testing and modeling is an iterative process that is expected to continue throughout the life of the program. 2. Laboratory Testing: The radiation shielding properties of candidate multifunctional spacecraft materials are characterized by means of measurements performed during controlled exposures at ground-based particle accelerators. Heavy ion exposures are made several times a year at the NASA Space Radiation Laboratory (NSRL) at the Brookhaven National Laboratory (BNL) and the Japanese National Institute for Radiological Sciences (NIRS) HIMAC heavy ion accelerator. Exposures are made to beams of elements found in relatively high abundance in the GCR spectrum such as H, He, C, Ne, Si and Fe in an energy range from 100 to 2000 MeV/nucleon. Materials are exposed to a simulated SPE using the Loma Linda University Medical Center (LLUMC) proton therapy synchrotron [Benton et al., 2002a]. Neutron exposures are made at the Los Alamos National Laboratory Neutron Science Center (LANSCE). During the exposures, samples of each shielding material are placed in the accelerator beamline and measurements are made behind the samples. Measurements are made using CR-39 Plastic Nuclear Track Detectors (PNTD), Thermoluminescent Detectors (TLD), Tissue Equivalent Proportional Counters (TEPC), and Liulin-4 Mobile Dosimetry Unit (MDU) portable Si spectrometers. LET spectrum, dose, and dose equivalent from particles of LET in water >5 keV/um is measured using CR-39 PNTD. Lineal Energy (y) Spectrum, dose and dose equivalent from particles of LET between 0.2 and 1000 keV/um is measured using the TEPCs. Dose and dose equivalent are measured by all the instruments. Measurements are made for a number of different thickness of each shielding material in order to determine how the shielding properties of the material change as a function of depth. 3. Computer Modeling: The radiation shielding properties of the candidate multifunctional spacecraft materials will be modeled using the analytic HZETRN and BRYNTRN, and the Monte Carlo MCNPX radiation transport codes. Modeling will be carried out both for the monoenergetic conditions used during Laboratory Testing and for the conditions encountered during interplanetary spaceflight. The model of Badhwar and O'Neill [Badhwar and O'Neill, 1992] will be used to determine the input GCR spectrum. SPE proton spectra will be modeled based on the CREME96 code [Adams, 1986; Tylka et al., 1997]. In modeling the monoenergetic exposures, quantities including the LET spectrum, charge spectrum, and y spectrum, in addition to dose and dose equivalent as functions of shielding depth, will be generated for comparison with the accelerator-based measurements. 4. Compiled Results and Database Implementation: Accelerator measurements and model results will be intercompared, ranked in order of shielding effectiveness, and compiled into a final report. A web-accessible database of spacecraft shielding materials will be established for use by NASA and the aerospace community. This database will be similar to materials and electronics parts databases currently maintained by various NASA centers. The database will contain not only information pertaining to the shielding effectiveness of each material, but also data on the physical properties of each material (e.g. tensile strength, thermal conductivity, etc.). The research program takes a pragmatic approach to identifying multifunctional spacecraft materals and characterizing their space radiation shielding properties. Candidate materials must possess both the potential to be effective in shielding space radiation and physical properties that make them suitable for use in spacecraft structures. For this reason, we are testing materials optimized for high hydrogen content, as well light weight materials (low mean atomic mass) currently in use or being developed by the aerospace community [Harris et al., 2002].
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Multifunctional materials developed and tested as part of this project may be of use in ground-based applications such as shielding of accelerators used in research and for cancer therapy. These materials may also one day be of use in reducing the radiation exposure of air crew aboard civilian and military aircraft.

Task Progress:	Candidate multifunctional materials tested include Carbon, Polyethylene, Kevlar, and fiberglass Composites, Martian Regolith Simulant (MRS), Lunar Regolith Simulant (LRS), MRS/epoxy composite, and LRS/epoxy composite. All materials were fabricated in house. Laboratory testing was carried out to thirteen separate ion/energy combinations at the BNL NSRL and AGS, and at the NIRS HIMAC. Thermal protection (heat shield) and multifunctional composite materials provided by NASA Ames Research Center were tested at NSRL to 1 GeV/n Fe, 1 GeV/n Si, 1 GeV/n O and 300 MeV/n Si. Data analysis of accelerator experiments conducted during 2004 was largely completed for the CR-39 PNTD exposures during the first half of 2005 and substantial progress was also made for the TEPC exposures. Processing and analysis of accelerator exposures made in 2005 are now underway for both CR-39 PNTD and TEPC.
Bibliography Type:	Description: (Last Updated: 11/13/2025)
Articles in Peer-reviewed Journals	DeWitt JM, Benton ER, Uchihori Y, Yasuda N, Benton EV, Frank AL. "Assessment of radiation shielding materials for protection of space crews using CR-39 plastic nuclear track detector." Radiat Meas. 2009 Oct-Nov;44(9-10):905-8. http://dx.doi.org/10.1016/j.radmeas.2009.10.041 , Oct-2009
Articles in Peer-reviewed Journals	DeWitt JM, Benton ER. "Secondary proton buildup in space radiation shielding." Life Sci Space Res. 2024 May;41:119-126. https://doi.org/10.1016/j.lssr.2024.02.005 ; PubMed PMID: 38670638 , May-2024
Presentation	Benton, E. R. "MMARSS: Multifunctional Materials Analysis of Radiation Shielding for Spacecraft" 16th Space Radiation Investigators' Workshop May-2005
Presentation	Benton, E. R. "Radiation Shielding for Tomorrows Spacecraft" Space Radiation Protection and Shielding, NIRS, Chiba, Japan Feb-2005
Presentation	Benton, E. R. "Radiation Shielding for Tomorrows Spacecraft" NASA Ames Research Center May-2005